Day 9: mathematical diagnostic

Name: _____ Date: _____

60 minutes total. No calculator, notes, internet or AI on the independent pass. Show your reasoning. If stuck, leave the useful work you have and move on. This is not a speed contest.

3 minutes setup · up to 30 minutes core · up to 20 minutes for two selected follow-ups · 7 minutes for your log. The tutor will supply follow-ups. Do not work through the whole reserve book.

If you finish a block early, check your work or continue to the next block with the tutor. Do not exceed the overall limit.

A5-3

Solve $4^x - 5 \cdot 2^x + 4 = 0$ over the reals.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

S5-3

On day k a machine produces $k \cdot 2^k$ parts. Find the total over days 1 through 6 without listing every individual part.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

C4-3

Ten people each shake hands with every other person once. How many handshakes occur? Explain why counting $10 \cdot 9$ needs correction.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

A5-2

Solve $2^{x+1}=16$ over the reals.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

S5-2

Find $1\cdot2+2\cdot2^2+3\cdot2^3+4\cdot2^4$.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

C4-2

How many shortest grid paths go from (0,0) to (3,2), using only one-unit right and up moves? Explain a correspondence.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

S5-1

Let $T=1\cdot r+2\cdot r^2+\cdots+n\cdot r^n$. Write $T-rT$ with like powers grouped. Do not assume $r\neq 1$.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

C4-1

What must you show to count one finite collection by matching it to another?

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

Session log

Which question was hardest to start, and what did you try?

_____

Was any question familiar, hinted, interrupted, or affected by outside study? Record IDs and details.

_____

Follow-up IDs attempted: _____ / _____